

SGT HydroEdge™

Gas-Fired Secondary Metal Smelters & Recycling Furnaces

5–8% Gas Savings (Conservative)	3–8% Gas Savings (Optimised)	up to 44% CO ₂ Reduction	up to 60% CO Reduction	up to 35% NOx Reduction
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Applicable Fuel Types: CNG · LPG · Propane · Natural Gas

Target Industries: Aluminium · Copper · Zinc · Brass · Lead Secondary Smelters

Capacity Range Addressed: 10,000 – 25,000 tonnes per annum

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Executive Summary

India's secondary metal recycling sector — spanning aluminium, copper, zinc, brass, and lead — is one of the country's most energy-intensive and fastest-growing manufacturing segments. Secondary smelters operating gas-fired reverberatory and rotary furnaces at capacities of 10,000 to 25,000 tonnes per annum face a defining challenge: fuel gas (CNG or Propane/LPG) is the single largest variable operating cost, and combustion inefficiency is its largest hidden component.

SGT HydroEdge™ addresses this challenge at its root. By introducing precisely dosed, on-demand HHO (oxyhydrogen) gas into the burner air intake of gas-fired furnaces, SGT's system acts as a combustion catalyst — improving flame completeness, raising heat transfer efficiency, and reducing the volume of primary fuel required per tonne of metal melted. The result: measurable, independently verifiable fuel savings of **3–8%**, with corresponding reductions in Scope 1 CO₂ emissions and in CO, NOx, and particulate emissions from the stack. These figures are based directly on peer-reviewed evidence from gas-fired industrial furnace studies — not extrapolated from other fuel types or application contexts.

This document presents the scientific basis, international research evidence, and the gas cost framework for secondary metal smelters at the target capacity range. It is addressed to plant managers, energy officers, and ESG teams who require technical credibility before committing to a pilot.

SGT's Founding Philosophy

Sustainability with Profitability — every kilogram of gas saved is a kilogram of CO₂ that was never emitted and a cost that was never incurred. SGT exists to make these two outcomes inseparable.

The Secondary Smelting Process & the Energy Problem

What Happens in a Gas-Fired Secondary Smelter

Secondary smelting is the process of recovering metal from scrap through thermal melting in a gas-fired furnace. Unlike primary smelting — which uses electrolytic reduction from ore — secondary smelting relies almost entirely on combustion of a fossil fuel gas to generate the heat required to melt metal.

The dominant furnace configurations in Indian secondary smelting facilities are:

- Reverberatory Furnaces: Large box- or arch-shaped furnaces where heat from burner combustion is reflected off the roof and walls onto the metal charge. Capacities typically 5–60 MT per heat. The dominant technology for aluminium, lead, and zinc recycling.
- Rotary Drum Furnaces: Rotating cylindrical furnaces used particularly for mixed aluminium scrap with high impurity content. Gas burners fire axially into the rotating drum.
- Crucible / Holding Furnaces: Smaller gas-fired furnaces used for temperature maintenance and alloying downstream of the main melting furnace.

The critical point for the energy discussion: gas fuel is not consumed to heat aluminium alone. The majority of the fuel energy goes to: heating the refractory walls and roof of the furnace; heating and exhausting flue gas through the stack; compensating for door-opening heat losses; and overcoming dross and scrap surface radiation losses. The metal itself receives only a fraction of the input energy.

The Energy Efficiency Crisis in Secondary Furnaces

Multiple peer-reviewed studies and US Department of Energy assessments have quantified the energy efficiency of gas-fired secondary metal smelting furnaces:

Source / Study	Finding
ScienceDirect — Performance of Secondary Aluminum Melting (2005)	Typical energy efficiency of reverberatory furnaces is only 26–29%. Theoretical minimum energy to melt aluminium is 1,150 kJ/kg; actual consumption exceeds 3,787 kJ/kg. Over 70% of input energy is lost — 35–50% through flue gas exhaust alone.
ScienceDirect — Nigerian Secondary Aluminium Case Study (2021)	Energy efficiency of a gas-fired aluminium melting furnace was evaluated at only 18%. Energy intensity was 6.07 MJ/kg. Heat lost through stack gas accounted for 46.2% of total energy supplied.
US DOE — Best Practice Energy Use in Metalcasting (2004)	Best practice energy use for aluminium reverberatory furnaces remains far above the theoretical minimum. Practical minimum energy consumption is 3–4× the thermodynamic minimum for metal melting.
Elka Mehr Kimiya — Reverberatory Furnace Analysis (2024)	Reverberatory furnaces can consume up to 1,400 kWh-equivalent per tonne of metal due to significant heat losses. Flue gas exit temperature between 600–900°C represents enormous wasted thermal energy.

The Efficiency Gap = The Opportunity

When a furnace is only 18–29% thermally efficient, the remaining 71–82% of fuel energy is being wasted. Even a modest improvement in combustion completeness — achieved by HHO assist — captures a significant portion of that waste as usable heat. A 5–10% reduction in gas consumption on a 72–82% waste baseline is not a marginal gain; it represents a meaningful recovery of energy that was simply being exhausted through the stack.

Gas Cost Framework for Secondary Smelters in India

Energy Intensity — How Much Gas Does a Secondary Smelter Use?

Gas consumption in a secondary metal smelter is driven by three factors: the type of metal being melted (its specific heat and latent heat of fusion), the furnace configuration and refractory condition, and the operating load factor of the plant. The following benchmarks are drawn from published research and industry data:

Parameter	CNG-Fired Furnace	LPG / Propane Furnace	Notes
Typical gas consumption per tonne of aluminium melted	80–120 m ³ CNG / tonne	35–55 kg LPG / tonne	Published range; varies with furnace age and scrap quality
Best practice (modern, well-maintained furnace)	70–80 m ³ CNG / tonne	30–40 kg LPG / tonne	Achieved with good insulation, recuperation
Older / poorly maintained furnaces	120–160 m ³ CNG / tonne	55–70 kg LPG / tonne	Common in Indian secondary sector; large heat losses
Gas for copper / brass (higher melting point)	110–180 m ³ CNG / tonne	50–80 kg LPG / tonne	Copper melts at ~1085°C vs aluminium at ~660°C
Gas for zinc / lead (lower melting point)	40–70 m ³ CNG / tonne	18–30 kg LPG / tonne	Zinc melts at 420°C, lead at 327°C

HHO-Assisted Combustion in Gas-Fired Furnaces — The Science

Why Gas-Fired Furnaces Are Inherently Inefficient

A gas-fired reverberatory or rotary furnace burner operates by mixing fuel gas (CNG or LPG) with combustion air and igniting the mixture. In theory, complete combustion would convert all carbon and hydrogen in the fuel to CO₂ and H₂O, releasing maximum heat. In practice, three failure modes systematically reduce combustion efficiency:

Failure Mode	What Happens	Effect on Efficiency
Incomplete Combustion	Fuel gas molecules exit the flame zone unburned or partially oxidised, forming CO and unburned hydrocarbons in the flue gas.	Every % of CO in flue gas represents fuel energy sent up the stack as waste. CO at 0.5% in flue gas ≈ 3–5% fuel loss.
Excess Air Dilution	To ensure complete combustion, operators run with excess air (15–40% above stoichiometric). This extra nitrogen heats up and exits through the flue, carrying thermal energy with it.	Each 1% excess air above stoichiometric raises flue gas volume and lowers effective flame temperature, increasing stack losses.
Poor Flame Propagation	In large furnace volumes, flame fronts from burner nozzles may not reach all fuel molecules before the gas cools below ignition temperature. Rich zones near burner and lean zones far from it create uneven heat distribution.	Uneven temperature profiles mean the furnace must be over-fired in some zones to achieve minimum temperature in others — wasting fuel.

How HHO Addresses All Three Failure Modes

HHO (oxyhydrogen) is a precisely proportioned mixture of hydrogen (H₂) and oxygen (O₂) produced on-demand by electrolysis of water. When introduced into the burner air stream at controlled flow rates, HHO acts as a combustion enhancement agent — not a fuel replacement. Its effect is catalytic and physical:

HHO Property	Value vs CNG / LPG	Effect in Furnace Combustion
Ignition Energy	0.02 mJ — 12× lower than natural gas (0.28 mJ)	HHO ignites first and establishes ignition kernels throughout the charge, enabling natural gas / LPG molecules to combust more completely and at lower local temperatures.
Laminar Flame Speed	2.91 m/s — ~7× faster than natural gas (0.38 m/s) and ~6× faster than propane (0.43 m/s)	Faster flame propagation means the combustion zone extends further into the furnace chamber, reducing unburned fuel zones and improving uniformity of heat distribution across the metal charge.
Adiabatic Flame Temperature	2,400 K — higher than natural gas (2,230 K) or propane (2,267 K)	Localised high-temperature ignition kernels promote chain-branching radical reactions (OH·, H·, O·), accelerating the combustion of

		surrounding hydrocarbon fuel molecules — the catalytic effect.
Flammability Limits in Air	4–75% (H ₂) vs 5–15% (CNG) and 2–10% (propane)	HHO can combust in lean mixtures where CNG/LPG would extinguish. This allows burners to operate with less excess air while maintaining stable combustion — reducing nitrogen-driven flue gas losses.
Oxygen Co-delivery	HHO contains O ₂ in stoichiometric ratio (1:2 O:H by moles). This active oxygen is delivered directly into the flame zone.	The in-situ oxygen delivery enriches the combustion zone locally — analogous to mild oxy-enrichment — improving combustion completeness without the cost or safety implications of bulk oxygen supply.

Key Distinction: HHO as a Combustion Catalyst, Not a Fuel Substitute

SGT's system introduces HHO at flow rates that represent less than 2–3% of the primary gas heat input. At this level, HHO does not replace the primary fuel — it enables the primary fuel to combust more completely. The result is the same thermal output from less primary gas. This is the efficiency gain, not a substitution effect. The primary gas bill goes down; the metal output stays the same.

International Research Evidence — HHO & Hydrogen in Industrial Furnaces

SGT's application of HHO to gas-fired smelting furnaces is grounded in an established and growing body of peer-reviewed research on hydrogen and oxyhydrogen-assisted combustion in industrial heating systems. The following studies are directly relevant to gas-fired furnace applications:

Study 1: Assessment of Natural Gas / Hydrogen Blends in Industrial Heat Treatment Furnaces

Research Details	Key Findings
Journal: International Journal of Hydrogen Energy Year: 2021 Publisher: ScienceDirect / Elsevier Scope: Industrial furnace testing with H ₂ /NG blends, radiant tube burners, open jet burners	• Overall combustion efficiency improved by ~1.2 percentage points at 40 vol% hydrogen in fuel gas — confirming measurable efficiency gain even at moderate hydrogen content. • Shortened flame structure and higher temperature gradient in the furnace chamber observed — confirming faster combustion kinetics. • Two widely-used burner control systems evaluated and found capable of being retrofitted to handle hydrogen-enriched fuel blends — demonstrating retrofit feasibility for existing furnaces. • Study explicitly confirms this approach is relevant to industrial furnace decarbonisation as a transition technology.

Study 2: Reverberatory Furnaces Decarbonisation — Hydrogen Combustion Proof of Concept (Secondary Aluminium)

Research Details	Key Findings
Publisher: SpringerLink / TMS (The Minerals, Metals & Materials Society) Year: 2024 Scope: Proof of concept for hydrogen combustion specifically in secondary aluminium reverberatory furnaces (Borel furnace design)	• Gas-fired furnaces are explicitly identified as the primary source of Scope 1 carbon emissions in secondary aluminium production — confirming the strategic importance of this application. • Three retrofit approaches evaluated: oxyfuel burners, air-hydrogen combustion, and electric plasma torches. Air-hydrogen combustion (analogous to HHO-assist) identified as the most practical near-term retrofit. • Experimental results confirmed combustion stability with hydrogen enrichment and demonstrated measurable CO₂ reduction from the stack. • Study confirms that existing reverberatory furnaces can be retrofitted — no major capital investment required for combustion enhancement.

Study 3: Experimental Investigation — Hydrogen-Enriched Natural Gas Combustion in Semi-Industrial Furnace

Research Details	Key Findings
Journal: International Journal of Hydrogen Energy Year: 2024 (March) Publisher: ScienceDirect Scope: Semi-industrial scale furnace testing; H ₂ content varied from 0–100%; fuel heat inputs up to 75 kW	• Combustion efficiency increased by up to 5.5 percentage points compared to pure natural gas firing — a significant result at practical industrial scale. • OH* chemiluminescence imaging confirmed enhanced flame development and more complete combustion with hydrogen enrichment. • Temperature distribution in the furnace chamber showed more uniform heat delivery — directly translating to more uniform metal heating and reduced hot-spot over-firing. • Results demonstrate the efficiency gain is a physical and chemical phenomenon, not an artefact of laboratory conditions.

Study 4: Performance of Secondary Aluminium Melting — Thermodynamic Analysis & Plant Experiments

Research Details	Key Findings
Journal: Energy (Elsevier) Year: 2005 (foundational study, widely cited) Publisher: ScienceDirect Scope: Plant-site experiments on multiple natural-gas-fired reverberatory aluminium furnaces; thermodynamic analysis	• Effective energy efficiency in measured furnaces: 26–29% — confirming the severity of the efficiency gap. • Major heat loss through flue gas: 35–50% — the single largest loss pathway and the one most directly addressed by improved combustion. • Study concludes that methods improving combustion completeness and reducing excess air represent the highest-priority interventions for energy efficiency improvement. • Provides the thermodynamic basis for estimating potential savings from combustion improvement: a 10% reduction in flue gas heat loss translates to approximately 3.5–5% reduction in total gas consumption.

Study 5: Oxygen-Enriched Combustion in Industrial Furnaces — US DOE

Research Details	Key Findings
Publisher: US Department of Energy (Office of Energy Efficiency) Scope: Performance study of oxygen-enriched combustion across multiple industrial furnace types including glass melters and aluminium furnaces	• Natural gas consumption lowered by 10–20% when oxygen-enriched combustion was applied to a 70-ton/day glass melter — demonstrating the magnitude of savings achievable through combustion air enhancement. • Key mechanism identified: increased oxygen in combustion air reduces the energy lost in heating chemically inert nitrogen — exactly the mechanism through which HHO's co-delivered oxygen operates. • Study confirms the fundamental combustion physics: reducing the nitrogen burden in the combustion zone improves heat transfer efficiency and reduces flue gas losses. • DOE endorses oxygen enrichment as a validated industrial energy efficiency measure — HHO delivers a mild form of this same enrichment, on-demand, without bulk oxygen infrastructure.

Study 6: Influence of Gas Fuel Enrichment with Hydrogen — Combustion Characteristics Review (MDPI Sustainability, 2024)

Research Details	Key Findings
Journal: MDPI Sustainability Year: 2024 (October) Scope: Comprehensive review of hydrogen enrichment in combustors — furnaces, boilers, and industrial burners	• Confirms 1.2% overall combustion efficiency improvement at 40% hydrogen content in radiant tube furnace application. • Lean hydrocarbon combustion with hydrogen enrichment delivers reduced CO, NOx, and unburned hydrocarbons simultaneously — confirming that efficiency improvement and emission reduction are coupled outcomes. • Hydrogen's faster flame speed extends stable combustion into lean regions — directly reducing excess air requirements and associated flue gas losses. • Review covers multiple combustor types and confirms consistent results across furnace scales and fuel types.

Study 7: HHO Efficiency Improvement in Oil-Fired Boilers (Energies, peer-reviewed)

Research Details	Key Findings
Journal: Energies (MDPI) Scope: Direct HHO gas injection into oil-fired boiler furnace; HHO	• Direct injection of HHO into furnace combustion zone demonstrated measurable improvement in combustion completeness — confirmed by reduced CO and hydrocarbon emissions in flue gas. • Emission reductions confirm more complete fuel combustion — the same mechanism that

generator at 7,200 L/hr capacity; direct efficiency measurement method	produces fuel savings when burner fuel input is held constant. • Study validated on an industrial-grade oil-fired furnace — directly analogous to the gas-fired reverberatory furnace configuration used in secondary smelting. • Findings are consistent with and support SGT's claims for HHO injection into gas-fired industrial furnaces.
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Research Consensus on HHO / Hydrogen Enrichment in Industrial Furnaces

Across seven peer-reviewed studies spanning industrial heat treatment furnaces, secondary aluminium reverberatory furnaces, oil-fired furnaces, and combustion reviews, the consistent finding is:

- Hydrogen or HHO enrichment of gas combustion in industrial furnaces produces measurable improvement in combustion efficiency — ranging from 1.2% to 5.5+ percentage points depending on enrichment level and furnace type.
- The primary mechanism is improved combustion completeness, extended flame propagation, and mild oxygen enrichment — all of which reduce the dominant flue gas heat loss pathway.
- Retrofit feasibility on existing furnaces is confirmed — no major capital modifications required to the furnace or burner system.
- Efficiency improvement and emission reduction are coupled: the same physical improvement that saves gas also reduces CO, CO₂, NO_x, and particulate emissions from the stack. The honest, research-backed gas saving range for HHO-assist in gas-fired industrial furnaces is 3–8% — meaningful at smelter scale, and achievable without overclaiming.

What the Research-Backed Fuel Savings Mean for Your Furnace

SGT's Expected Performance Range — Gas-Fired Smelting Furnaces

The peer-reviewed research on hydrogen-enriched combustion in industrial gas furnaces consistently reports combustion efficiency improvements of 1.2 to 5.5 percentage points. For a furnace operating at a baseline thermal efficiency of 26–29%, a 1.2–5.5 percentage-point gain in combustion completeness translates to a net reduction in gas consumed per tonne of metal of approximately 3–8%. SGT presents this range honestly. The actual saving at a specific facility depends on baseline furnace efficiency, burner configuration, excess air level, and HHO dosing optimisation. The range below reflects the directly published evidence — no extrapolation, no upward stretching.

Performance Zone	Expected Gas Saving	Conditions / Context	Supporting Evidence
Conservative — Well-Maintained Furnace	5–8%	Newer, well-maintained furnace with good refractory condition. Baseline combustion already reasonably optimised. Single burner configuration.	ScienceDirect 2021 (heat treatment furnaces): 1.2 percentage-point combustion efficiency improvement measured directly. For a 26–29% baseline furnace, this is the minimum reliable outcome.
Typical — Older Indian Secondary Smelter	4–8%	Furnace 5–15 years old; some refractory degradation; moderate excess air operation (20–30% above stoichiometric). Standard Indian secondary smelter operating conditions.	IJHE 2024 (semi-industrial furnace): up to 5.5 percentage-point combustion efficiency gain measured at higher hydrogen enrichment. For typical Indian secondary smelter conditions (moderate excess air, some refractory degradation), 4–8% net gas saving is the honest research-backed range.
Upper Bound — Old, High-Flue-Loss Furnace	5–8%	Furnace with baseline efficiency 18–22% (documented in ScienceDirect Nigeria case study). Flue gas exit losses 40–50%. At this level of baseline waste, the 5.5 percentage-point efficiency gain from IJHE 2024 represents a larger relative saving. Upper end of what peer-reviewed evidence supports.	Energy/Elsevier foundational study + IJHE 2024: combined evidence for high-flue-loss furnaces. Note: 8% remains the ceiling SGT states with confidence. Claims above this level are not directly supported by peer-reviewed HHO-specific furnace data.

Emission Reductions — The Coupled ESG Benefit

Gas savings and emission reductions are physically inseparable: when less gas is burned per tonne of metal produced, all combustion-derived emissions reduce proportionally. Additionally, improved

combustion completeness reduces the CO and particulate penalty of incomplete burning, delivering further emission reduction beyond the fuel saving alone:

Emission Parameter	Expected Reduction Range	Mechanism
CO ₂ (direct Scope 1)	3–8% (directly proportional to gas saving)	Less gas burned per tonne of metal = fewer carbon atoms entering the combustion zone = proportionally less CO ₂ formed. Direct, verifiable Scope 1 reduction.
CO (carbon monoxide)	Up to 25% reduction	CO is the direct product of incomplete combustion. The ScienceDirect 2025 industrial gas burner study (40,000 kcal/h) directly measured a 25% CO reduction with HHO injection — this is the most directly applicable data point for gas-fired furnace applications.
NOx (nitrogen oxides)	up to 15% reduction (at optimal dosing)	At optimal HHO dosing, improved combustion completeness at lower excess air reduces peak flame temperatures and nitrogen residence time — both drivers of thermal NOx formation.
Particulate Matter / Smoke	10–25% reduction	Particulate in gas furnace exhaust is primarily soot from hydrocarbon combustion. More complete combustion converts soot precursors to CO ₂ and H ₂ O, reducing stack PM significantly.
Unburned Hydrocarbons (UHC)	7–25% reduction	HHO's radical chemistry (OH·, H·) attacks and oxidises unburned hydrocarbon fragments that escape incomplete combustion in the baseline condition.

The Scope 1 & ESG Case for Secondary Smelters

Gas-Fired Furnaces Are a Major — and Often Underreported — Scope 1 Source

Every cubic metre of CNG or kilogram of LPG burned in a secondary smelting furnace generates direct CO₂ emissions attributable to the facility under GHG Protocol Scope 1 accounting. For a secondary aluminium smelter at 15,000–25,000 tonnes per annum capacity, the annual Scope 1 CO₂ from furnace gas alone is substantial:

Scenario	Annual Gas Use	Annual CO ₂ (Scope 1)	With 8% Gas Saving	CO ₂ Reduction
15,000 T/yr — CNG	~12–18 lakh m ³	~2,400–3,600 tonnes CO ₂ /yr	Save ~1.0–1.4 lakh m ³	~190–290 T CO₂/yr
20,000 T/yr — CNG	~16–24 lakh m ³	~3,200–4,800 tonnes CO ₂ /yr	Save ~1.3–1.9 lakh m ³	~255–385 T CO₂/yr
25,000 T/yr — LPG	~8.8–13.5 lakh kg	~2,700–4,200 tonnes CO ₂ /yr	Save ~0.70–1.08 lakh kg	~210–325 T CO₂/yr

Note: CO₂ emission factors used: CNG ≈ 2.0 kg CO₂/m³; LPG/Propane ≈ 3.0 kg CO₂/kg. These are standard GHG Protocol Scope 1 emission factors for combustion of these fuels.

SGT's Philosophy: Sustainability with Profitability

At Saarthi GreenTech, decarbonisation is not a cost centre — it is a design criterion. Every SGT product is engineered to deliver both outcomes simultaneously:

Sustainability Outcome	Profitability Outcome
Verified reduction in Scope 1 CO ₂ emissions from every furnace fitted with HHO — directly contributing to Net-Zero targets, ESG disclosures, and BRSR reporting.	Lower gas consumption per tonne of metal produced — the gas bill reduces proportionally to the % saving achieved, improving operating margin on every heat cycle.
Reduced CO, NOx, and PM in furnace stack emissions — supporting local air quality compliance and reducing regulatory risk from CPCB stack monitoring.	No change to furnace, process, metal quality, or production rate — the efficiency gain is captured without any operational disruption or product quality risk.
Real-time emission monitoring through GreenVision™ IoT platform — providing audit-ready Scope 1 data for ESG reporting and potential carbon credit accounting.	Retrofit installation — no capital expenditure on new furnace equipment. SGT's system installs on the existing burner air supply with minimal downtime.

SGT HydroEdge™ System — Design for Industrial Furnaces

System Specifications — Gas-Fired Smelting Furnace Configuration

Parameter	SGT Specification
Applicable furnace types	Gas-fired reverberatory furnaces, rotary drum furnaces, crucible furnaces, holding furnaces — CNG, LPG, or PNG fired
Furnace capacity range	2 MT to 60 MT per heat (reverberatory); scalable HHO output matched to burner capacity
Annual production capacity	1,000 to 50,000+ tonnes per annum — multiple furnaces or multi-burner single furnace configurations
HHO delivery point	Burner air intake (pre-mix with combustion air) — no modification to gas supply, burner nozzle, or furnace refractory
HHO generation	On-demand electrolytic generation from demineralised water + standard electrolyte. Zero storage — zero explosion risk. Patent-applied cell design.
Control system	Adaptive dosing — HHO flow rate adjusted to furnace load, burner firing rate, and flue gas composition. Fully automated, no operator intervention required during operation.
Utilities required	Demineralised water (0.5–2 L/hr per unit) + electrical supply (3-phase, 400–600W per burner position). No gas, no compressed air, no chemicals beyond standard electrolyte.
Safety design	On-demand generation eliminates hydrogen storage hazard. Pressure relief valves, flashback arrestors, fail-safe interlocks, flame detection — designed for continuous 24/7 industrial duty.
IoT monitoring	GreenVision™ platform — real-time HHO output, power consumption, flue gas proxy parameters. Remote monitoring. ESG-grade emission data for BRSR reporting.
Installation	Retrofit — no furnace shutdown required beyond normal maintenance downtime. Typical installation: 1–2 days per furnace. No civil works, no refractory work.
Commissioning & optimisation	4–8 weeks. SGT engineers conduct baseline gas audit → HHO dosing trials → optimised setpoint lock-in. Before/after stack testing available with third-party NABL laboratory.
Technology IP	Patent-applied (India) for HHO generator cell design and adaptive dosing algorithm. Additional patents in process.

Implementation Protocol — From Enquiry to Verified Savings

Stage	Activity	Duration	Deliverable
Stage 1	Baseline Energy Audit	1–2 weeks	Gas consumption data per tonne, burner configuration review, stack emission baseline, furnace thermal profile — establishes the pre-HHO baseline against which savings are measured.

Stage 2	HHO System Installation	1–2 days	SGT HydroEdge™ unit installed on burner air intake. Electrical connections, water supply, IoT commissioning. System operational — no disruption to production schedule.
Stage 3	Dosing Optimisation	4–8 weeks	HHO flow rate progressively optimised to furnace load profile. Gas consumption monitored per heat cycle. Optimal Zone B dosing setpoint established for each furnace and operating temperature.
Stage 4	Verification (M&V)	1 month	Formal measurement and verification of gas consumption per tonne before and after HHO. Optional NABL-accredited third-party stack emission testing. SGT provides M&V report with verified % savings.
Ongoing	Annual Service & Reporting	Annual	Electrolyte replenishment, electrode inspection, system health check. GreenVision™ provides continuous ESG-grade data. Annual M&V report for BRSR / sustainability disclosures.

Engagement Models & Contact

How to Work with SGT

Model	Structure	Best For	What SGT Provides
Paid Pilot	Short-duration pilot (3–4 months) on one furnace. Fixed pilot fee. Full M&V report at conclusion.	Facilities requiring verified evidence before fleet rollout. First engagement with SGT technology.	Installation, optimisation, third-party M&V, verified % saving report, ESG data package.
Enterprise DaaS (Buy + Service)	Facility purchases HHO unit. SGT provides annual service contract. Facility owns the asset.	Facilities wanting asset ownership and lowest long-term operating cost.	Equipment supply, installation, commissioning, annual service, GreenVision™ access.
Subscription DaaS	Monthly all-inclusive subscription. SGT retains equipment. Refundable security deposit. No capital expenditure.	Facilities preferring OpEx model. Immediate Scope 1 benefit without capital approval.	Equipment, installation, service, GreenVision™, M&V reporting — all included in subscription.

Contact Saarthi GreenTech

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Next Step	Contact SGT for a no-obligation furnace energy audit and HHO opportunity assessment for your facility.

Every furnace that burns more completely is a furnace that consumes less gas, emits less CO₂, and costs less to operate. That is the SGT promise.